Hitesh Kumar

Software Engineer 2 at MathWorks

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EDUCATION

National Institute of Technology Kurukshetra (NITK), M.Tech in Computer Engineering, CGPA: **9.26**/10.0 — Aug 2022 – Jul 2024

AKTU University (GL Bajaj), B.Tech in Computer Science & Engineering, CGPA: **8.96**/10.0 — Aug 2018 – Jul 2022

WORK EXPERIENCE

Software Engineer 2, MathWorks Pvt. Ltd. — May 2024 – Present

AuthNZ & Connection Manager | C++ | MATLAB | Google Benchmark | CppMicroServices | Perforce

- **Designed and implemented** a Connection Interface and supporting Connection Manager methods, enabling management of cloud service provider connections using the **Factory design pattern**.
- **Architected** a mock Connection Provider as a **CppMicroServices** bundle, decoupling connection flows from live cloud credentials (S3, Google Sheets) for **deterministic, isolated** integration testing.
- **Engineered** a **Google Benchmark** suite across **14** Connection Manager APIs at Core and SPF (IPC) layers; **diagnosed and resolved** a **110%** latency regression in `listSecrets`.

MATLAB Interface to C++ | C++ | MATLAB | clibgen | Perforce

- **Implemented** `setInputArgument`, `setProperty`, `setOutput`, and `excludeMethod` APIs powering **C++ interface scripting workflows**, extending support for C++ classes of varying types and sizes.

GenAI Developer Tooling | Python | LLMs | RAG | MCP | Docker

- **Developed AI agents** automating developer workflows (code review, test triage, changelist analysis) by orchestrating LLMs over MathWorks tooling via the **Model Context Protocol (MCP)**.
- **Built** a **RAG** pipeline over internal code and docs with **vector embeddings** and a **multi-agent** architecture of specialized **subagents** for parallel fan-out, improving relevance and throughput.

Multimedia | C++ | MATLAB | Perforce

- **Developed** class-based test suites for image (**JPEG**, **PNG**) and VideoReader (**MPEG**, **AVI**) APIs, achieving **98%** code coverage.

Co-op Intern, AMD Bengaluru | Python | Grafana | MYSQL | Perforce — July 2023 – April 2024

- **Automated** the **end-to-end** Debug Tracker Dashboard for Regression and Simulation, **reducing** manual effort by **98%** and accelerating bug resolution via automated owner notifications.
- **Reduced** average signature-request response time to **900 ms** by capping JSON payloads at **1 MB** and optimizing post-processing.

Software Engineer, MAQ Software, Noida | PowerBI | MYSQL | DAX | Git — Oct 2021 – Aug 2022

- **Developed** interactive Power BI dashboards for course-purchase analytics, **improving** report load time to **2 seconds** by filtering redundant data and optimizing fact-dimension table relationships.

PROJECTS

Scalable URL Shortener | Java | Spring Boot | Redis | PostgreSQL | Docker | AWS | Kafka — Github

- **Architected** a **distributed**, horizontally scalable URL-shortening service with **Spring Boot REST APIs**, achieving sub-**100 ms** p99 latency at **10K+** req/s.
- **Designed** a **fault-tolerant** system with **consistent hashing** for sharding, **Redis** caching, **Kafka** async analytics, and **rate limiting**; deployed on **AWS** (ECS, ELB) via **Docker** at **99.9%** availability.

The Book Spot | Python | Django | JavaScript | SQLite — Github

- **Developed** an online e-book store with **Payment Gateway**, **Face Authentication**, **Text-to-Speech**, **RESTful** endpoints, **session-based auth**, and a purchase-history **recommendation** module.

RESEARCH PAPER CONTRIBUTION

- **Best Paper Award** – improved **mAP**/**mIoU** by **5%** on a cross-attention transformer for query-based video-moment detection. *Connecting Words to Motion*, Springer Nature.
- **Best Paper Award** – proposed a language-guided multimodal **Cross-Attention Transformer** (**CLIP ViT-B/32**) for video summarization, gaining **3% F1**/**BLEU** on **TVSum**/**SumMe** with **zero-shot** generalization. *Advancing Video Summarization*, **IEEE IConSCEPT 2024**.

SKILLS

- **Languages**: C, C++, Java, Python, SQL, JavaScript, MATLAB
- **Concepts**: OOD, Design Patterns, System Design, Distributed Systems, DSA, Multithreading, IPC, REST APIs
- **GenAI**: LLMs, RAG, AI Agents, Multi-Agent Systems, MCP, Vector Embeddings, Prompt Engineering
- **Frameworks & Cloud**: Spring Boot, NodeJS, Django, TensorFlow, AWS, Docker, Kafka, Redis, PostgreSQL, Git, Perforce, CI/CD

ACHIEVEMENTS AND EXTRACURRICULAR ACTIVITIES

- Secured **AIR 2113** amongst **0.1 Million** students in **GATE CS 2023**; solved **600+** problems on LeetCode and GeeksforGeeks.